

Module 2

Test Planning & Strategy in Depth (SV/UVM)

Learning objectives

- Turn your initial verification plan into a detailed, reviewable **test strategy**, with a structure...

Prerequisites

- See module README

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

flowchart LR

T1["Test Taxonomy and Structure"]

T2["Test Catalogue Refinement"]

T1 --> T2

T3["Negative, Stress, and Corner"]

T2 --> T3

Turn your initial verification plan into a detailed, reviewable **test strategy**, with a structured test catalogue, negative/stress planni...

Overview

- Module 1 gave you a high-level verification plan, scope, and early test catalogue. In this module,...

How to learn this module

Suggested learning path (1/2)

- Re-read ``module1/VERIFICATION_PLAN.md`` and ``module1/REQUIREMENTS_MATRIX.md``.
- Scaffold Module 2 workspace if needed:
``./scripts/module2.sh --scaffold``
- Fill ``module2/TEST_PLAN.md``, ``REGRESSION_PLAN.md``, and optionally refine ``COVERAGE_PLAN.md``.
- Map each test intent to a future UVM test name, sequence, and regression tier.
- Validate completeness: ``./scripts/module2.sh --check``

Suggested learning path (2/2)

- Complete `module2/CHECKLIST.md` before Module 3.

Design architecture

1. UVM environment skeleton

- ``common_dut/tb/stream_fifo_env_skeleton.sv`` — env + agent stubs (source/sink agents).
- Hierarchy: ``uvm_test`` → ``stream_fifo_env`` → agents → (future) driver, sequencer, monitor.
- Interfaces tie to DUT ports via virtual interfaces or pin-level connections in the top testbench.

Rtl Architecture

```
Diagram: Rtl Architecture
(render with mmdc when available)
flowchart TB
    subgraph DUT["stream_fifo (common_dut/rtl)"]
        SRC["Source interface\ns_valid / s_ready / s_data"]
        MEM["Circular buffer\nmem[DEPTH], wr_ptr, rd_ptr"]
        SNK["Sink interface\nm_valid / m_ready / m_data"]
        STS["Status\nlevel, overflow, underflow"]
    end
```

2. Test-plan ↔ TB mapping

- `TEST\_PLAN.md` catalogue entries map to UVM tests, sequences, and config knobs.
- `REGRESSION\_PLAN.md` defines tiers that will eventually select subsets of tests.
- Naming conventions (`SMK\_`, `FTR\_`, `ERR\_`) link documentation to future `uvm\_test` names.

```
Verification Architecture  
  
Diagram: Verification Architecture  
(render with mmdc when available)  
flowchart TB  
  subgraph TB["UVM testbench (planned)"]  
    ENV["stream_fifo_env"]  
    A_S["source agent(driver, seq, mon)"]  
    A_M["sink agent"]  
    ENV --> A_S
```

3. Test execution mapping (planning level)

- Each catalogue row specifies: UVM test class, base sequence, plusargs/seeds, and regression tier.
- Smoke tests map to short directed sequences; stress tests map to long-run or backpressure scenarios.
- Error-injection tests document illegal handshake or overflow/underflow stimulus intent.
- ``./scripts/module2.sh --check`` confirms test plan sections and TB skeleton references exist.

Handshake logic (push/pop)

```
assign s_ready = (count < DEPTH);  
    assign m_valid = (count != 0);  
    assign level    = count;  
  
    // Determine when to push/pop  
    always_comb begin  
        push = s_valid && s_ready;  
        pop  = m_valid && m_ready;  
    end
```

Agent stub — build_phase

```
class stream_fifo_agent_skeleton extends uvm_agent;

    `uvm_component_utils(stream_fifo_agent_skeleton)

    function new(string name = "stream_fifo_agent_skeleton",
uvm_component
parent = null);
        super.new(name, parent);
    endfunction

    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        // Stub: add driver, sequencer, monitor in later modules
    endfunction

    function void connect_phase(uvm_phase phase);
        super.connect_phase(phase);
        // Stub: connect driver to sequencer, etc
```

Key files to study

Open these in the repo

- ``module2/TEST_PLAN.md`` — detailed test catalogue and taxonomy
- ``module2/REGRESSION_PLAN.md`` — tier definitions and pass/fail policy
- ``module2/COVERAGE_PLAN.md`` — coverage intent linked to tests
- ``common_dut/tb/stream_fifo_env_skeleton.sv`` — env hierarchy your tests will target
- ``common_dut/rtl/stream_fifo.sv`` — handshake and status behavior to test
- ``scripts/module2.sh`` — structure and completeness checks

Verification & testing methods

1. Test taxonomy

- Smoke/sanity — fast checks after every change; feature — per-requirement depth.
- Stress / long-run — backlog pressure, sustained traffic; error injection — overflow, underflow, ill...
- Each type documents expected runtime class and whether seeds are fixed or varied.

Testing Methods

Diagram: Testing Methods

(render with mmdc when available)

flowchart LR

SMK["Smoke / sanity"] --> CORE["Core feature"]

CORE --> STR["Stress / long-run"]

STR --> ERR["Error injection"]

ERR --> REG["Regression tiers/sanity → nightly → full"]

REG --> META["Test metadata: \nID, seed, runtime class"]

2. Regression tiering (planning level)

- Assign every catalogue test to sanity, core, stress, or full tiers.
- Document pass/fail policy preview and which negative tests are safe for frequent runs.
- ``./scripts/module2.sh --check`` verifies plan completeness and TB skeleton presence.

3. Reproducibility

- Plan seed strategy, constraint overrides, and plusargs per test for debuggable random runs.

Module 2 — scaffold and check

```
#!/usr/bin/env bash

# Module 2: Test Planning & Strategy Orchestrator
# This script summarizes and checks the planning artifacts for Module
# 2.
# Usage: ./scripts/module2.sh [OPTIONS]

set -euo pipefail

# Colors for output
RED='\033[0;31m'
GREEN='\033[0;32m'
YELLOW='\033[1;33m'
```

Syllabus topics

1. Test Taxonomy and Structure (1/2)

- Test Types
- Smoke/sanity vs feature vs stress vs error-injection vs performance (if applicable).
- Directed vs constrained-random vs scenario-based tests.
- Tiers and Frequency
- Per-commit / pre-push tests.
- Nightly regression suites.

1. Test Taxonomy and Structure (2/2)

- Weekly full or long-running suites.
- Mapping to UVM
- How test types relate to UVM tests and sequences.
- Using configuration objects / plusargs / factory overrides to specialize tests.

2. Test Catalogue Refinement (1/2)

- From List to Catalogue
- Expanding the initial 10–20 test intents into a more complete catalogue.
- Grouping tests by feature, interface, or requirement group.
- Naming and Metadata
- Test ID conventions (e.g., `SMK\_`, `FTR\_`, `STR\_`, `ERR\_` prefixes).
- Metadata: owner, runtime class (short/medium/long), deterministic vs random, seed handling.

2. Test Catalogue Refinement (2/2)

- Redundancy and Value
- Identifying overlapping tests and merging where reasonable.
- Distinguishing “coverage-driving” tests vs “regression guardrail” tests.

3. Negative, Stress, and Corner-Case Planning (1/2)

- Negative Testing
- Illegal sequences, protocol violations, out-of-range values.
- Malformed packets/transactions, timing violations (where meaningful).
- Stress and Long-Run Scenarios
- High-load / long-duration tests.
- Pseudo-random traffic, back-to-back operations, resource exhaustion.

3. Negative, Stress, and Corner-Case Planning (2/2)

- Corner Cases
- Boundary values, wrap-around conditions, reset-in-the-middle, mode transitions.
- Instrumentation Needs
- What additional checks, monitors, or assertions are needed to support these tests.

4. Seeds, Constraints, and Reproducibility (1/2)

- Seed Strategy
- When to fix seeds vs vary them.
- How to record seeds for debugging and reproducibility.
- Constraints and Randomization Control
- High-level constraints in sequences vs constraint overrides per test.
- Guarding against “over-constrained” tests that hide bugs.

4. Seeds, Constraints, and Reproducibility (2/2)

- Test Parameters and Configurations
- Using UVM configuration database, plusargs, or config files.
- Planning for parametrization (e.g., data width, depth, mode).

5. Regression Tiering and Policies (Planning Level) (1/2)

- Regression Tiers
- Tier definitions (e.g., `sanity`, `core\_feature`, `stress`, `full`).
- Target runtimes and approximate test counts for each tier.
- Test Assignment
- Mapping each test in the catalogue to one or more tiers.
- Deciding which negative/stress tests are safe for frequent runs.

5. Regression Tiering and Policies (Planning Level) (2/2)

- Pass/Fail Policies (Preview)
- What counts as a pass, fail, flake.
- Basic quarantine strategy for flaky tests (details in later modules).

6. Coordination with Coverage and Requirements

- Closing Loops with Requirements
- Ensuring each high-priority requirement has enough tests (not just one).
- Verifying that tests exercise planned coverage points.
- Coverage-Driven Test Gaps
- Identifying missing tests from coverage perspective (e.g., unhit bins).
- Planning targeted tests to close meaningful coverage holes.

Command reference highlights

Scaffold and validate Module 2

- `./scripts/module2.sh --scaffold`

Review test and regression templates

- `head -40 module2/templates/TEST_PLAN.md`

Hands-on examples

Module 2 self-check

```
$ ./scripts/module2.sh --help
```

Terminal: module2_check

```
[[0;36m=====[[0m
[[0;36mModule 2: Test Planning & Strategy in Depth[[0m
[[0;36m=====[[0m
```

Usage: ./scripts/module2.sh [OPTIONS]

Module 2: Test Planning & Strategy in Depth (SV/UVM)
This script summarizes and checks the planning artifacts for Module 2.

OPTIONS:

--test-plan-status	Show status of test_plan.md only
--regression-status	Show status of regression_plan.md only
--coverage-status	Show status of coverage_plan.md only
--checklist-status	Show status of checklist_module2.md only
--structure-status	Show required-sections check only
--summary	Show all statuses (default)
--check	Media/CI self-check (structure only)
--demo	Print example commands from EXAMPLES.md
--scaffold	Copy templates/*.md into module2/ if missing
--help, -h	Show this help message

EXAMPLES:

```
# Full summary
./scripts/module2.sh

# Focus only on test plan status
./scripts/module2.sh --test-plan-status

# Focus only on regression plan
./scripts/module2.sh --regression-status
```

Exercise scaffold

```
./scripts/module2.sh --scaffold
```

Demo: Test plan template

```
$ head -30 module2/templates/TEST_PLAN.md
```

Terminal: ex01_test_plan

Detailed Test Plan (Module 2)

> **Purpose**: Refine and expand the test catalogue and test strategy for your DUT.

> **Module**: 2 – Test Planning & Strategy in Depth (SV/UVM)

> **Instructions**: Fill in this document for your design. Copy from `module2/templates/` if nee

1. Context

<!-- TODO: Project name, DUT name, link to Module 1 artifacts. Summarize key decisions from Modu

2. Test Taxonomy

2.1 Test Types

<!-- TODO: Define test types (smoke, feature, stress, error, performance). -->

2.2 Test Tiers

<!-- TODO: Define tiers (e.g. SANITY, CORE, STRESS, FULL) with cadence and target runtime. -->

3. Test Metadata and Conventions

3.1 Naming and IDs

<!-- TODO: Test ID prefixes, UVM test class names, sequence naming. -->

3.2 Runtime and Stability

<!-- TODO: Runtime class, determinism, flake risk. -->

Demo: Regression plan template

```
$ head -25 module2/templates/REGRESSION_PLAN.md
```

Terminal: ex02_regression_plan

Regression Plan (Module 2)

> **Purpose**: Define initial regression tiers, policies, and test-to-tier mapping.

> **Module**: 2 – Test Planning & Strategy in Depth (SV/UVM)

> **Instructions**: Fill in this document for your design. Copy from `module2/templates/` if nee

1. Regression Goals

<!-- TODO: Primary goals (detect regressions, runtime targets, stability). -->

2. Regression Tiers

<!-- TODO: Table of Tier, Purpose, Contents, Cadence, Target Runtime. -->

3. Test Assignment to Tiers

<!-- TODO: Map test IDs from test_plan.md to tiers. -->

4. Runtime Budgeting

<!-- TODO: Per-tier runtime estimates and totals. -->

5. Pass/Fail, Flakes, and Quarantine (Planning-Level)

Demo: Coverage plan template

```
$ head -25 module2/templates/COVERAGE_PLAN.md
```

Terminal: ex03_coverage_plan

Coverage Plan Refinement (Module 2)

> **Purpose**: Refine your coverage model and connect it more tightly to tests and regression.

> **Module**: 2 – Test Planning & Strategy in Depth (SV/UVM)

> **Instructions**: Fill in this document for your design. Copy from `module2/templates/` if nee

1. Context

<!-- TODO: Link to module1 verification plan. What this document adds. -->

2. Functional Coverage Model

2.1 Covergroups and Placement

<!-- TODO: Table of Coverage ID, Location, Purpose. -->

2.2 Coverpoints and Crosses

<!-- TODO: For each covergroup, describe coverpoints and crosses. -->

3. Code Coverage Expectations (Planning)

<!-- TODO: Statement/branch/toggle goals, known exclusions. -->

Demo: Reference test plan

```
$ head -20 module2/.solutions/TEST_PLAN.md
```

Terminal: ex04_solutions

Detailed Test Plan (Module 2)

> **Purpose**: Refine and expand the test catalogue and test strategy for your DUT.
> **Module**: 2 – Test Planning & Strategy in Depth (SV/UVM)

1. Context

- **Project name**: Stream FIFO Verification
- **Block/DUT name**: `stream\_fifo`
- **Module 1 artifacts**:
 - Verification plan: `module1/VERIFICATION\_PLAN.md`
 - Requirements matrix: `module1/REQUIREMENTS\_MATRIX.md`

Summarize any key decisions from Module 1 that affect test planning:

- DUT is a parameterizable streaming FIFO with ready/valid handshakes, supporting DATA_WIDTH=8 a
- Three core requirements identified: R1 (in-order transfer), R2 (overflow flag), R3 (underflow
- Initial test catalogue of 12 tests (T1-T12) established covering smoke, feature, stress, and e
- Coverage model planned with CG_LEVEL, CG_OPS, CG_FLAGS, and CROSS_LEVEL_FLAGS.
- Scoreboard (SB_MAIN) and SVA assertions (A_OVERFLOW, A_UNDERFLOW) planned for checking.

Demo: Module validation

```
$ ./scripts/module2.sh --check
```

Terminal: ex05_validate

```
Module 2: Test Planning & Strategy in Depth
Module 2: media self-check
```

```
OK: module2/ exists
OK: templates/ exists
OK: EXAMPLES.md exists
OK: docs/MODULE2.md exists
OK: common_dut/rtl/ exists
```

All required checks passed.

Practice & assessment

What you should know (1/6)

- Design a structured test taxonomy tailored to your DUT and UVM environment.
- Maintain a rich test catalogue with clear metadata and priorities.
- Plan explicit negative, stress, and corner-case tests.
- Define initial regression tiers and assign tests to them.
- Explain how your test plan supports requirements coverage and coverage closure.
- In ``module2/TEST_PLAN.md``, define:

What you should know (2/6)

- Test types with definitions and examples.
- Tiers (sanity, core, stress, full) with expected runtime budgets.
- For at least 10 existing tests from Module 1, assign:
 - Type, tier, and expected runtime.
- Grow your catalogue to 25–40 tests (intents), focusing on:
 - Feature coverage.

What you should know (3/6)

- Interface/protocol scenarios.
- Mode and configuration combinations.
- Identify at least 3 redundant or low-value tests and:
 - Merge or remove them, documenting the decision.
 - Add at least:
 - 3 negative tests (illegal, malformed, or out-of-range behavior).

What you should know (4/6)

- 3 stress or long-run tests.
- For each, define:
- Preconditions and exact stimuli.
- Expected behavior (including assertions/flags).
- Special logging or debug support needed.
- Decide:

What you should know (5/6)

- Which tests will run with fixed vs varying seeds.
- How seeds will be passed (e.g., plusargs, config).
- Document this in `module2/TEST\_PLAN.md` and update `module1/REQUIREMENTS\_MATRIX.md` if needed.
- In `module2/REGRESSION\_PLAN.md`, define:
 - Tier names, cadence (per-commit, nightly, weekly), and runtime targets.
- Inclusion criteria for tests in each tier.

What you should know (6/6)

- For each test in your catalogue, add a column indicating:
- Primary tier(s) it belongs to.

Exercises

- Runtime Budgeting
- Flake-Resistance Design
- Traceability Check

Assessment checklist

- Has a written test taxonomy and tier definition.
- Test catalogue expanded and de-duplicated with clear metadata.
- Negative, stress, and corner-case tests are explicitly planned.
- Seed and randomness policy is documented.
- A draft regression plan exists with test-to-tier mapping.
- Requirements and coverage implications are reflected in the updated plan/matrix.

Summary & next steps

- Turn your initial verification plan into a detailed, reviewable ****test strategy****, with a structure...
- Complete module2/CHECKLIST.md
- Review module2/EXAMPLES.md and run each lab
- Next: Coverage Planning & Analysis in Practice